

Summary

M.Sc. Data Science student specializing in NLP, Deep Learning, and ML. Skilled in Python, Transformers, RAG, and AI Agents. Passionate about building scalable, LLM-driven systems.

Education

M.Sc. Data Science & AI (SGPA: 8.80) **2024 – 2026**

Ramakrishna Mission Residential College Narendrapur (Affiliated to University of Calcutta)

B.Sc. Statistics (CGPA: 9.10) **2021 – 2024**

Bidhannagar College

Technical Skills

Languages & Data: Python, R, SQL, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI

ML/DL: Regression, Classification, CNN, LSTM, Transformers, LLMs, GenAI, GANs

RAG & Agents: RAG, MMR, BM25, LangChain, LangGraph, LlamaIndex, FAISS, AstraDB, Multi-Agent, Prompt Engineering

MLOps & Tools: MLflow, DVC, Docker, AWS EC2, FastAPI, Streamlit, Git, Azure

Experience

Associate Software Consultant Intern (AI/ML, TuTeck Technologies) **2026**

- Built a **Hybrid RAG + Text-to-SQL** HR chatbot combining PDFPlumber context injection with LLM-generated SQL over a **CSV → SQLite** pipeline for client deployment.
- Integrated **Groq LLaMA-3** into a two-source retrieval architecture, grounding all responses in real data and eliminating hallucinations on factual HR queries.

Research Intern – Machine Intelligence Unit, ISI Kolkata **2025**

- Analyzed **10K+ colorectal cancer records** using survival analysis for recurrence forecasting.
- Engineered **ML-driven chart system** for cancer image/PDF detection, boosting processing efficiency by **30%**; cut data prep time by **40%**.

FastAPI File Management Intern, IDEAS (ISI Kolkata) **2025**

- Built FastAPI service for CSV/JSON upload, merging, and persistence using PostgreSQL.
- Implemented UUID-based cache + error handling, reducing API latency by **35%**.

Projects

MarketMind (Multi-Agent Financial Analysis) — CrewAI, FastAPI, GPT-4o-mini, Azure, Firecrawl [GitHub]

- Orchestrated **2 CrewAI agents** across **4 specialized tools** to generate BUY/SELL/HOLD recommendations, streaming live output via **SSE** with **yfinance** and **Firecrawl** integration.
- Reduced manual research time by **~70%**; persisted reports to **Azure Blob Storage** and logged **100%** of analyses to **Azure PostgreSQL** for full history tracking.

SafeSpace — AI Mental Health Therapist — MedGemma, LangGraph, FastAPI, Streamlit, Twilio [GitHub]

- Built an agentic AI therapist using **Google MedGemma** with a **LangGraph** workflow across **3 tool-use decisions** (Twilio emergency calling, location services, professional referral).
- Deployed full-stack via FastAPI + Streamlit; reduced crisis response initiation time by **~60%** vs manual triage.

LoanIQ — Two-Way Loan Approval & Amount Prediction — Scikit-learn, Streamlit [GitHub] | [Live Demo]

- Designed a **two-stage ML pipeline** (classification + regression) using Random Forests achieving **~90%+** approval accuracy across **10+ financial features**; deployed live on Streamlit.

Blinkit Sales Analytics Dashboard — Power BI, DAX, Power Query [GitHub]

- Built a **Power BI** dashboard across **3 dimensions** (outlet size, location, type) tracking **4 KPIs**; authored **DAX measures** and **Power Query** pipelines cutting reporting time by **~50%**.

Vision-to-Language Image Captioning — CNN-LSTM, VGG16, Transfer Learning [GitHub]

- Built a **CNN-LSTM encoder-decoder** using **VGG16** transfer learning trained on **8K+ image-caption pairs**, achieving a **BLEU-4 score of 0.58** with grammatically coherent outputs.
- Reduced training time by **~40%** via transfer learning vs training from scratch; automated image tagging for scalable content pipelines.

Certifications

- IBM SkillsBuild – *Agentic AI: From Learner to Builder (AI Agent Architect)*, Jul–Aug 2025
- Deloitte – Data Analytics Job Simulation (*Forage*), Jul 2025 | **The Complete SQL Bootcamp (Udemy)**, 2025
- HP LIFE – *Data Science & Analytics*, Nov 2024

Achievements

- 2nd Place – Analysis 2025 (Data Visualization), St. Xavier's College | Football Champion – Inter-departmental Tournament